Frozen-self-energy charge response and the projector contribution to LAPW stress

Implementation and validation note for DFT+eDMFT

8 November 2026

Purpose. Obtain the strain derivative of the Kohn–Sham potential without two converged strained calculations and without an impurity solve. The immediate application is the chain-rule contribution from the potential-dependent, normalized, finite-band-window projector. This note distinguishes a tested complete charge-map derivative from the less accurate stored-band approximation. The correction remains opt-in.

1 What is differentiated, and what is kept fixed?

Let η be hydrostatic linear strain, with deformation gradient $F_\eta = (1+\eta)I$, determinant $J_\eta = (1+\eta)^3$, and reference cell volume Ω. Fractional nuclear positions, muffin-tin radii, local orbital axes, impurity self-energy on its supplied Matsubara mesh, and the double-counting contribution contained in that input are held fixed in the present tests. Band-window endpoints are fixed integer indices.

Use material coordinates $\mathbf{x}$ in the central cell:

$$q_\eta(\mathbf{x}) = J_\eta n_\eta(F_\eta \mathbf{x}), \qquad v_\eta(\mathbf{x}) = V_{\mathrm{KS},\eta}(F_\eta \mathbf{x}). \tag{1}$$

Here n is the physical total electron density, including core charge; V_{KS} is the local Kohn–Sham potential. A subscript 1 denotes $\partial_\eta|_0$. Equation (1) gives

$$q_1 = n_1 + 3n_0 + \mathbf{x}\cdot\nabla n_0, \qquad v_1 = V_1 + \mathbf{x}\cdot\nabla V_0. \tag{2}$$

The derivative V_1 on the right is at fixed physical position; v_1 is at fixed material position. Confusing the two loses the advection term. For general infinitesimal strain E, replace 3 by $\operatorname{Tr} E$ and $\mathbf{x}\cdot\nabla$ by $(E\mathbf{x})\cdot\nabla$.

The first implementation constrains total particle number. A Mott-insulator calculation at fixed chemical potential is a different response problem; the complete WIEN driver does not silently substitute fixed N for that ensemble. The analytic band-space action exposes both choices.

2 The linear screening equation

Write the linearized density-to-potential map in material coordinates as

$$v_1 = v_{\mathrm{aff}} + K q_1. \tag{3}$$

v_{aff} is the potential derivative when $q_\eta = q_0$: nuclei and cell are strained, while physical density is transported affinely. K is the central Hartree plus exchange-correlation kernel. In the numerical implementation LAPW0 evaluates its action, including nonspherical, interstitial and GGA contributions; no explicit dense kernel is built.

The frozen-self-energy electronic response satisfies

$$q_1 = b + \chi_{\rm fr} v_1, \qquad s = b + \chi_{\rm fr} v_{\rm aff}, \qquad \boxed{(I - \chi_{\rm fr} K) q_1 = s.} \tag{4}$$

b is the explicit geometrical response of the one-electron charge map, including basis and reconstruction changes, at fixed material potential. The source s is obtained together, so its strongly cancelling pieces need not be evaluated separately. One screened approximation is $q_1 \simeq s$, followed by $v_1 \simeq v_{\rm aff} + Ks$. It is not sufficient in these tests.

Why this is not a full interacting DMFT susceptibility

In a fixed central orthonormal representation,

$$G_k(z) = [z + \mu - H_k - U_k \widetilde{\Sigma}(z) U_k^\dagger]^{-1}, \tag{5}$$

$$\delta G_k = G_k \left[\delta H_k - \delta\mu I + \delta U_k \widetilde{\Sigma} U_k^\dagger + U_k \widetilde{\Sigma} \delta U_k^\dagger \right] G_k. \tag{6}$$

$z = \mathrm{i}\omega_n$, $\widetilde{\Sigma}$ is the local self-energy after the implemented double counting, and U_k embeds local correlated orbitals. There is no $U_k \delta\widetilde{\Sigma}_{\rm imp} U_k^\dagger$ in this frozen-Σ calculation. The physical DMFT susceptibility would require that term and the impurity vertex. The screening in Eq. (4) uses the DFT kernel K, not an assumed Hubbard-U RPA kernel.

Computing δG here supplies q_1 and hence v_1. It does not mean adding an independent δG term to a stationary functional derivative. Stationarity at fixed projectors does not remove the explicit chain rule when the implemented projector itself depends on $V_{\rm KS}$.

3 Finite window and normalization must both respond

Let ϕ collect the untruncated localized trial orbitals, and let $\Pi_{W,k}$ project onto the chosen KS bands at each k. Then

$$U_k = \Pi_{W,k} \phi_k X, \qquad S = \mathrm{Sym} \sum_k w_k \phi_k^\dagger \Pi_{W,k} \phi_k, \qquad X = S^{-1/2}. \tag{7}$$

The operation Sym is the actual orbital-symmetry and equivalent-site average used by the code. The k weights w_k are constant and sum to one. For several impurity blocks these expressions apply with the implemented block and site bookkeeping. Inactive orbital channels retain the code's identity normalization; they must not acquire an artificial derivative.

For a perturbing potential matrix D_k in the central KS basis, and indices $n \in W$, $a \notin W$,

$$(\delta\Pi_{W,k})_{an} = \frac{(D_k)_{an}}{\epsilon_{kn} - \epsilon_{ka}}, \qquad (\delta\Pi_{W,k})_{na} = (\delta\Pi_{W,k})^*_{an}. \tag{8}$$

Within-W and within-complement blocks vanish for this spectral-projector derivative. A window boundary through a degeneracy is rejected, not regularized. Smooth fixed-index selection still requires an isolated window subspace.

In the frozen central LAPW basis,

$$\delta S = \mathrm{Sym} \sum_k w_k \phi_k^\dagger \delta\Pi_{W,k} \phi_k, \tag{9}$$

$$\delta U_k = \delta\Pi_{W,k} \phi_k X + \Pi_{W,k} \phi_k \delta X. \tag{10}$$

If $S = V\,\mathrm{diag}(s_i) V^\dagger$, the exact Fréchet derivative is

$$(V^\dagger \delta X V)_{ij} = -\frac{(V^\dagger \delta S V)_{ij}}{\sqrt{s_i}\sqrt{s_j}(\sqrt{s_i} + \sqrt{s_j})}. \tag{11}$$

No projector pseudoinverse is needed: the exported X is authoritative. The saved normalization can differ slightly from the newly summed S; directional endpoint tests anchor to the saved central X consistently.

The stored-band implementation uses only available complement bands in Eq. (8). It also holds central radial augmentation fixed. The complete charge-map derivative below includes the response missing from both restrictions, while keeping the same physical DMFT window.

4 Matsubara summation and particle-number constraint

For one spin component define $H_{\infty,k} = H_k + U_k \widetilde{\Sigma}_\infty U_k^\dagger - \mu I$, $G_{\infty,k}(z) = (z - H_{\infty,k})^{-1}$, and $f_T(x) = (e^{x/T} + 1)^{-1}$. The implemented density prescription is

$$\rho_k = g_s w_k \left[f_T(H_{\infty,k}) + 2T \sum_{n \in \mathcal{M}} a_n \,\mathrm{Herm}\{G_k(\mathrm{i}\omega_n) - G_{\infty,k}(\mathrm{i}\omega_n)\} \right]. \tag{12}$$

$g_s = 2$ in the tested nonmagnetic calculation. $\mathcal{M}$ is the kept positive Matsubara mesh and a_n are the actual production quadrature weights. The native export supplies them, T, μ, and the effective self-energy matrices in Ry. The action differentiates Eq. (12), including the static subtraction and exact Fermi-matrix derivative. It does not square a quadrature-weighted Green function: a_n appears once after the two resolvents in Eq. (6) are multiplied.

In an eigenbasis of H_∞, the static derivative is

$$[L_f(H_\infty; \delta H_\infty)]_{ij} = f_T[\varepsilon_i, \varepsilon_j](\delta H_\infty)_{ij}, \quad f_T[x,y] = \frac{f_T(x) - f_T(y)}{x - y}, \quad f_T[x,x] = f_T'(x). \tag{13}$$

Let $\delta\rho^{(0)}$ be the fixed-μ response and $\rho^{(\mu)}$ the response to a unit increase in μ. Fixed total particle number gives

$$\delta\mu = -\frac{\sum_k \mathrm{Tr}\, \delta\rho_k^{(0)}}{\sum_k \mathrm{Tr}\, \rho_k^{(\mu)}}, \qquad \delta\rho_k = \delta\rho_k^{(0)} + \delta\mu\, \rho_k^{(\mu)}. \tag{14}$$

A nearly zero denominator requires an explicit ensemble decision. The analytic action refuses to divide by it. A uniform potential shift at fixed N gives $\delta U = 0$, $\delta\mu = \delta V$, and $\delta\rho = 0$.

5 Complete directional derivative without an impurity solve

Define $\mathcal{F}_\eta[q]$ as *one* frozen-Σ charge evaluation, pulled back to the reference cell. Each evaluation consists of LAPW0, LAPW1, dmft2 with refreshed normalization and chemical potential, lcore, and the standard interstitial charge normalization. No mixer history or impurity solve is used. Then

$$s \simeq \frac{\mathcal{F}_{+h}[q_0] - \mathcal{F}_{-h}[q_0]}{h_+ - h_-}, \tag{15}$$

$$Ax \equiv \chi_{\mathrm{fr}} K x \simeq \frac{\mathcal{F}_0[q_0 + tx] - \mathcal{F}_0[q_0]}{t}. \tag{16}$$

Actual rounded lattice strains $h_\pm$ are used. The central baseline is recomputed with the same program path and normalization as each action. This avoids dividing a small central mismatch by t.

The second line is a numerical action of a linear-response operator at the central cell, not a converged strained calculation. It includes radial response and the complete LAPW eigenvector response implicitly: computing the occupied/window eigenvectors at the perturbed potential lets all basis states participate without writing every unoccupied band.

Plain mixing can diverge. With residual $r_j = s + Aq_j - q_j$, the implementation uses Anderson history:

$$\gamma = \arg\min_{\gamma} \|r_j - \Delta R\gamma\|_2, \qquad q_{j+1} = q_j + \beta r_j - (\Delta Q + \beta \Delta R)\gamma. \tag{17}$$

Here ΔQ and ΔR contain differences of recent iterates and residuals. Only this small history problem uses SVD. The returned iterate must have a measured residual below tolerance; otherwise the driver exits with a nonconvergence status. Tolerances refer to the stored density-vector norm, not to a universal physical norm; stress and step convergence are checked independently.

LAPW-specific safeguards

- The source and response runs use exactly the central integer G set, through the existing serial WIEN fixed-G reference markers. Fixed band count alone does not ensure a differentiable basis.
- Real versus complex LAPW1 is selected from its unit-5 definition, not from the mere presence of an `in1c` file. Empty optional files can exist in real-vector cases.
- The spherical `clmval` channel contains $\sqrt{4\pi}\, r^2 n$, whereas `clmsum`/`clmcor` contain $4\pi r^2 n$. Only the $l = m = 0$ valence channel needs the extra $\sqrt{4\pi}$ factor before adding core density. Higher harmonics are unchanged.
- For radial density coefficients $c(r) = r^2 n_{lm}(r)$, affine transport is $c_\lambda(r) = c_0(r/\lambda)/\lambda$; pullback is $\lambda c_\lambda(\lambda r)$. Fourier density star sums acquire λ^{-3} and λ^3, respectively. Cubic interpolation on the logarithmic radial mesh is used. Boundary extrapolation is small but must be checked by halving h; density at R_{MT} is not assumed zero.
- The analytic signed density matrix is reconstructed through the existing dmft2 radial/angular/FFT machinery. All signed eigenmodes are retained. The input includes eigenvectors to reject a mismatched gauge, including rotations within degenerate subspaces. Energies printed during this reconstruction are not response energies.

6 Projector stress contraction and its limits

Let R_k^W and R_k^X be the already implemented adjoint matrices for the window and normalization chain rules, defined operationally by

$$\delta\Gamma^V_{\mathrm{proj}} = \Re \sum_k \mathrm{Tr}[(R_k^W + R_k^X)\, \delta v_k^{\mathrm{op}}]. \tag{18}$$

Here δv_k^{op} is the matrix of the *material* local-potential derivative in the central KS representation. The exported adjoints already contain the code's spin, frequency, and k weights; none is applied again. The correction to mean tensile stress is

$$\boxed{\Delta\bar{\sigma} = \frac{1}{3\Omega}\Re \sum_k \mathrm{Tr}[(R_k^W + R_k^X)(v_{\mathrm{aff}} + Kq_1)_k^{\mathrm{op}}].} \tag{19}$$

Multiply Ry/Bohr^3 by 14710.507641078 for the GPa conversion used in these comparison archives. Compressive pressure has the opposite sign. The tabulated values are already hydrostatic means; do not divide by three again. For tetragonal FeSe this is the trace divided by three, not any individual diagonal component.

The diagnostic contraction uses a frozen-central-basis LAPW1 matrix probe and a complete central dual basis, with separate MT advection and interstitial material-coordinate treatment. This probe is a

reference integrator; it does not imply that the density response can freeze its radial basis. The general production replacement would evaluate those potential matrix elements directly.

The driver writes predicted physical endpoint potentials using $n_\pm(\mathbf{r}) = J_\pm^{-1}[q_0 \pm hq_1](F_\pm^{-1}\mathbf{r})$ and two LAPW0 calls. Subtracting the raw `vns` arrays alone is not the material potential operator: sphere/interstitial and advection terms must still be treated consistently.

7 Validation results

All results below are mean tensile stresses in GPa, using saved frozen-Σ finite-difference Klein references from the preceding audit. Those reference totals were not recomputed by an impurity solver during this work. Neither their densities nor potentials were inputs to the response solver; they were used only for independent validation.

Material	Previous analytic	Corrected	Numerical	Difference
$SrVO_3$	+5.676884	+2.690161	+2.933836	-0.243675
FeSe	-0.116492	-1.118667	-1.049151	-0.069516

Material	Predicted correction	Relaxed reference	Difference	Actions
$SrVO_3$	-2.98672306	-2.98675885	+3.58e-05	9
FeSe	-1.00217579	-1.00221547	+3.97e-05	11

The final validation used a fixed integer G set, fixed band counts (43 in $SrVO_3$, 66 in FeSe), and fixed windows $[12, 31]$ and $[21, 44]$, respectively. The retained meshes have 75 and 140 k points. The source strain was $h = 2.5 \times 10^{-4}$ and the central density-probe step $t = 10^{-5}$. The screened-density residual norms were 0.01213 and 0.00893. Their differences from the relaxed-reference material-density response were 0.01504 and 0.01352, versus reference norms 1062.99 and 997.37.

Halving both steps to $h = 1.25 \times 10^{-4}$, $t = 5 \times 10^{-6}$ gives:

Material	Correction, half steps	Change
$SrVO_3$	-2.98672795	-4.88e-06
FeSe	-1.00219128	-1.55e-05

What exposed the incomplete approximation?

Insert the *known* relaxed-reference density derivative into the right-hand side of Eq. (4); measure the residual against that reference. Using the corrected fixed-G source, the density-vector norms are:

Charge response used	$SrVO_3$	FeSe
Stored bands, frozen radial basis	170.264 (43 bands)	426.406 (66 bands)
100 stored bands, frozen radial basis	109.944	—
200 stored bands, frozen radial basis	89.013	—
Complete LAPW response, frozen radial basis	75.731	—
Complete LAPW and radial response	0.0174	0.0562

The complete frozen-radial test freezes the basis in *both* LAPW1 and dmft2 reconstruction. Freezing only LAPW1 would mix inconsistent radial and coefficient representations. The results establish that adding a modest number of complement bands alone does not fix the charge-response action. Radial-basis response is significant in $SrVO_3$.

Checks passed

- The exported-mesh Python density reproduces the production central band density to 3.3×10^{-15} in $SrVO_3$ and 9×10^{-16} in FeSe.
- Independent nonlinear spectral-projector endpoint differences verify Eq. (6) and the normalization derivative with quadratic step convergence. A small standalone algebra fixture tests the same identity.
- Fixed-N conservation and uniform-potential gauge tests pass.
- Signed density reconstruction retains negative modes; an intentionally changed eigenvector gauge is rejected with a failing process status.
- With optional hooks disabled, $SrVO_3$ charge output is identical to the prior result, and FeSe differs by at most 1.1×10^{-18} in the stored density coefficients.
- LAPW0 applied to the measured reference density derivative reproduces the independently measured potential correction to approximately 2×10^{-5} GPa ($SrVO_3$) and 5×10^{-5} GPa (FeSe).
- Final fixed-G source and action runs preserve the G sets, band counts, k ordering and isolated window boundaries. Early preliminary runs that allowed G changes are superseded by these results.

The screening actions took about 169 seconds in $SrVO_3$ and 138 seconds in FeSe on this laptop, with four MPI ranks for dmft2 and serial LAPW1. Summed program times including source and central evaluations were about 218 and 172 seconds, before the separate stress matrix probe. These are measured diagnostic costs, not a general scaling guarantee. There were no impurity solves and no pairs of converged strained charge calculations.

Remaining error. The total-stress residuals of 0.244 and 0.070 GPa are not explained by the numerical accuracy of this potential-response calculation. That does not identify their source: residual analytic basis approximations, the complement convergence of the stress adjoint itself, and consistency of the functional finite-difference reference still need separate checks. Agreement of the new correction with its reference must not be reported as exact agreement of the complete stress.

8 Files and practical use

The reusable sources are in `src/python/`:

- `frozen_sigma_response.py`: analytic fixed-basis stored-band action, including projector/window/normalization and Matsubara derivatives.
- `frozen_sigma_screening.py`: matrix-free Anderson solver.
- `wien_response_density.py`: density conventions and transport.
- `wien_frozen_sigma_response.py`: complete WIEN charge-map action, source construction, screening and potential output.
- `frozen_sigma_projector_stress.py`: guarded hydrostatic contraction of the saved potential and projector-adjoint matrices.

Native hooks are in `src/dmft2/frozen_response_io.f90`, with optional calls from `l2main.f90`. They are disabled unless requested. A separate `dmft2_response` executable is installed; the ordinary installed dmft2 is not replaced by an automatic stress correction.

The complete driver takes a JSON configuration. Its workspace must be new. It copies inputs and never modifies the reference calculation. A template is included beside this note. Activate the usual environment, then run:

```
source ~/activate
~/dbin/wien_frozen_sigma_response.py response_config.json
```

The configuration names the reference case, workspace, WIEN root and executable, MPI launcher, and LAPW1/dmft2 definition files. Definitions must use relative case paths; the driver rewrites the distributed-vector suffix for the serial LAPW1 output. The initial complete driver supports nonmagnetic primitive orthogonal cells with fixed N and projector 5. It does not claim magnetic, SOC, shear, or general-cell validation.

For the analytic action, first replay a central TOT dmft2 charge calculation with recomputeEF=0 and the saved converged chemical potential:

```
export EDMFT_RESPONSE_EXPORT=/absolute/work/central
mpirun -n 4 ~/dbin/dmft2_response dmft2.def
unset EDMFT_RESPONSE_EXPORT
~/dbin/frozen_sigma_response.py --export /absolute/work/central \
  --potential dV_band.npy --output /absolute/work/density_response
export EDMFT_RESPONSE_DENSITY_PREFIX=/absolute/work/density_response
mpirun -n 4 ~/dbin/dmft2_response dmft2.def
unset EDMFT_RESPONSE_DENSITY_PREFIX
```

Run the last command in an isolated copy: its `clmval` is a signed response density and must not replace an ordinary converged density. The matrix input has shape (N_k, N_b, N_b) and units Ry per strain, in exactly the exported central eigenvector gauge. N_b includes all stored bands.

9 What remains open

The tested response can reproduce the potential-dependent projector correction obtained from the saved relaxed endpoints. It does not by itself prove that every remaining term in the stress functional is exact. The residual total-stress errors remain explicit in the table. Their audit should use the same ensemble, normalized projector, band/G sets, and frozen-Σ Klein comparison including its impurity-functional linear correction. No values from MnO or magnetic MnTe are inferred from these nonmagnetic metallic tests.

For speed, the next production step is an analytic radial response and a complete complement action replacing the numerical charge-map tangent. The stored-band implementation provides a useful building block and an independent check, but the measured radial error rules out promoting it unchanged. The present complete tangent is a validated, moderately priced reference route for deriving that faster implementation.